

Course syllabus for

Academic year 2022/2023 >

FTF141 - Thermodynamics and statistical mechanics

Termodynamik och statistisk mekanik

Course syllabus adopted 2022-02-10 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57148

Open for exchange students: No

Block schedule: [D+](#)

Maximum participants: 180

Only students with the course round in the programme overview

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0122 Examination 7,5 c Grading: TH	7,5 c						Sat 22/10-2022 am J, Wed 04/01-2023 am J, Wed 16/08-2023 am J

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 3 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)

Examiner:

[Henrik Grönbeck](#)



[Go to course homepage](#)

Theme:

Environment 1,5 hec

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Classical mechanics and basic quantum physics. Multivariabel analysis. Basic probability theory and statistics.

Aim

The purpose of the course is to provide basic knowledge about thermodynamic relations and statistical distributions, so that the student after finished course will be able to apply these laws to various practical problems and also be able to assimilate new scientific and technical information in order to attack novel and more complicated technological problems.

Learning outcomes (after completion of the course the student should be able to)

- describe the concept of energy in thermodynamics and distinguish the concepts of energy, heat and work;
- perform thermodynamic calculations for different processes and understand the significance of quasi-static processes;
- explain the concept of entropy based on a microscopic description;
- calculate entropy and other thermodynamic quantities based on a microscopic description of simple model systems;

- explain the relationship between the microscopic description of statistical mechanics and the phenomenological thermodynamics;
- describe and apply the first and second laws of thermodynamics for closed and open systems;
- explain the concept of free energy and apply this in connection with thermodynamic equilibrium and available work;
- explain the basics of phase equilibrium, both in one- and two-component systems;
- explain the basic distributions in statistical mechanics and be able to use them in different applications;
- explain black body radiation and set up relationships for radiation balance;
- explain quantum statistics and be able to apply this to ideal quantum gases and master the concept of state density;
- with thermodynamic concepts analyze energy balances and energy utilization in relation to sustainable development and the environment

Content

Basic thermodynamic concepts as equilibrium, reversible and irreversible processes, state functions, heat and work. Statistical description of many-body systems and the concept of multiplicity and entropy. The laws of thermodynamics. Application of thermodynamics to heat engines, refrigerators and heat pumps. Thermodynamic potentials, free energies and chemical potential. Phase equilibrium in one and two-component systems. Microcanonical, canonical and grand canonical ensembles. Equipartition theorem. Maxwell velocity distribution. Applications to classical ideal gases, lattice vibrations, paramagnetism and adsorption problems. Fermi-Dirac and Bose-Einstein distribution functions for ideal quantum gases and density of states. Applications to electrons in metals and in semi-conductors. Planck distribution, blackbody radiation and heat balance applied to the earth atmosphere.

Organisation

Instruction is given in the form of lectures and problem-solving sessions. These are supplemented by one compulsory experimental exercise.

Literature

Daniel V. Schroeder: An Introduction to Thermal Physics (Oxford University Press).

Examination including compulsory elements

The course ends with a written exam including questions that probe conceptual understanding and problem solving skills. During the course there is two home assignments, which may give points for the exam. The experimental exercise is compulsory.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.